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Research paper

Physics-informed operator learning with dual-critic proximal policy optimization for time-sensitive PDE-constrained control: Application to river pollution management

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ABSTRACT

Physics-informed operator learning offers spatially resolved surrogate prediction with governing-equation guidance but does not alone ensure operationally feasible control. In contrast, reinforcement learning-based methods achieve computational scalability through policies that may exploit simulator artifacts and generalize poorly outside their training regimes. A physics-informed operator learning framework is proposed that integrates deep operator networks with a dual-critic proximal policy optimization architecture for rapid constrained control. The key innovation combines a one-step physics-informed Deep Operator Network surrogate trained by supervised learning and physics-consistency regularization, achieving an untouched test relative L_2 error of 2.21% and a defect-corrected advection-diffusion-reaction root mean square error of 0.7077, with a dual-critic architecture in which an environmental critic learns from simulator feedback while a separate physics-diagnostic critic provides surrogate-derived advection-diffusion-reaction consistency guidance. Comprehensive validation spans simulator fidelity assessment, surrogate accuracy characterization, controller ablations and held-out generalization testing across unseen parameter regimes and operational disturbances. Relative to standard proximal policy optimization, the physics-informed dual-critic controller increased point-time compliance from 94.20% to 94.86%, reduced integrated exceedance by 11.65% and shortened violation duration by 10.85% at statistically comparable treatment effort; hierarchical 95% confidence intervals supported all three safety differences. The physics-informed controller retained improvements across the evaluated out-of-distribution and operational-stress conditions, while the data-only and physics-informed controllers remained statistically comparable. These results support separated environmental and surrogate-derived diagnostic value estimation within the synthetic advection-diffusion-reaction benchmark.

1. Introduction

River pollution represents a critical environmental challenge affecting freshwater ecosystems, public health, and aquatic resource management across industrialized and developing regions worldwide [1]. The transport, mixing, and transformation of aquatic contaminants, including agricultural runoff, industrial effluents, and municipal wastewater, follow deterministic physical principles governed by advection, diffusion, and reaction kinetics [2,3]. Predicting how pollutant concentrations evolve spatially and temporally is essential for designing effective intervention strategies and supporting water quality compliance assessment. Simulation plays a foundational role by enabling environmental managers to forecast pollution trajectories, evaluate control interventions, and assess long-term management outcomes without the prohibitive costs and ethical concerns of real-world experimentation [4].

Time-sensitive decision-making for pollution control demands the ability to rapidly simulate system dynamics and respond to pollution events within operationally feasible timescales. Traditional simulation methods based on explicit numerical solutions of governing partial differential equations are computationally expensive, often requiring seconds to minutes per forward evaluation at sufficient spatial and temporal resolution [5]. For environmental systems where optimal intervention timescales are comparable to forward simulation timescales, this computational burden can become prohibitive for time-sensitive control

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optimization. Moreover, uncertainty quantification for understanding how predictions vary under parameter uncertainty and stochastic environmental perturbations typically requires ensemble simulations that multiply computational cost further [6]. Thus, simulation efficiency is inextricably linked to control feasibility: the faster we can predict pollution dynamics, the more rapidly we can evaluate and optimize control actions [4].

The spatiotemporal dynamics of pollutant concentration in riverine systems are fundamentally governed by partial differential equations (PDEs) coupling advection, diffusion, and reaction mechanisms [3]. Numerical simulation of advection-diffusion-reaction (ADR) equations requires careful spatial discretization to ensure stability and accuracy across diverse parameter regimes [2]. Temporal integration employs explicit schemes to maintain computational tractability, while boundary conditions and source/sink terms are enforced to represent realistic riverine configurations [7]. Environmental stochasticity is incorporated through spatially correlated Gaussian increments sampled independently across numerical time steps, reflecting unmodeled processes and measurement uncertainties [6]. The complete simulator provides a testbed for exploring pollution trajectories, testing control strategies, and assessing physical consistency before higher-fidelity or field-calibrated studies [8].

Traditional environmental control approaches employ relatively straightforward strategies: predetermined rules based on historical data, manual interventions triggered by monitoring threshold exceedances, or periodic compliance checks [4]. While straightforward to implement, these approaches are fundamentally reactive, responding only after violations occur, and inflexible, unable to anticipate or adapt to dynamic conditions. Classical environmental management has historically employed static zoning and fixed emissions standards that set predetermined pollution limits at sources without dynamic optimization. Feedback control via distributed sensor networks places monitoring stations downstream and adjusts valve settings based on observed violations, but this approach is inherently reactive and cannot anticipate pollution events. Offline trajectory optimization computes control sequences in advance without feedback, making this approach brittle to model mismatch and environmental disturbances. Expert-designed heuristics are interpretable but suboptimal, require manual redesign, and do not leverage data for improvement. Traditional approaches face fundamental limitations, including computational scalability, model dependency, and inability to guarantee constraint satisfaction under stochastic realizations [6]. The optimality-responsiveness tradeoff forces slow methods to sacrifice real-time responsiveness for guaranteed optimality, while fast methods sacrifice optimality through oversimplification. Traditional control does not leverage historical data or adapt to changing environmental conditions.

Reinforcement learning (RL) methods have emerged as promising alternatives for learning control policies directly from data or simulation without requiring explicit mathematical models [9,10]. Policy gradient methods, such as Proximal Policy Optimization (PPO) and Trust Region Policy Optimization (TRPO), learn control policies by interacting with an environment and iteratively improving the policy through gradient ascent on expected cumulative reward [11]. Actor-critic architectures combine a policy network that selects actions with value function networks that estimate expected reward, enabling low-variance policy gradient estimation and improved training stability [12–14]. Policy gradient methods naturally generalize to high-dimensional action spaces and handle stochastic environments, making them well-suited for distributed environmental control applications [15,16].

Despite these advantages, standard RL methods treat underlying system dynamics as black-box functions, ignoring domain-specific knowledge about physical processes. RL typically requires millions of environment interactions to learn effective policies; in real river systems with slow timescales, collecting this many high fidelity samples is impractical [17]. RL policies tend to exploit quirks or biases in the simulator rather than learning generalizable principles, leading to poor performance upon deployment. RL policies are black-box neural networks whose decisions lack transparency, which regulators and environmental managers distrust [18]. Standard RL provides no guarantees that learned policies will satisfy safety or regulatory constraints, resulting in potential violations during exploration or testing. Policies trained on specific parameter regimes often fail to generalize to new regimes, requiring costly retraining. RL convergence is notoriously unstable, particularly with function approximation and stochastic environments, leading to convergence to poor local optima or divergence.

Scientific Machine Learning (SciML) integrates domain knowledge, such as governing PDEs, conservation laws, and boundary conditions, into machine learning model architectures and training objectives to mitigate RL limitations. Physics-Informed Neural Networks (PINNs) encode partial differential equation (PDE) constraints as additional loss terms during training, ensuring predictions respect physical laws [19,20]. PINNs have achieved success in solving inverse PDE problems and generating surrogate models with improved sample efficiency and generalization [21]. Physics-Informed (PI) approaches provide strong regularizers that dramatically reduce data requirements, enabling models trained with PDE-constraints to generalize well to Out-Of-Distribution (OOD) parameters and initial conditions [22]. These approaches ensure outputs respect conservation laws and boundary conditions by construction [23].

However, SciML approaches face significant limitations for real-time control [18]. PINN loss landscapes are notoriously difficult to optimize, requiring careful tuning for balancing data loss and PDE residual loss, misbalancing leads to slow convergence, local minima, or constraint violations [17]. PINNs excel at solving fixed initial-value problems but struggle to learn operators (mappings from initial conditions and parameters to solution fields) in high-dimensional function spaces of time-evolving PDEs required for control applications. While PINN inference is fast, training involves second-order automatic differentiation at many collocation points, resulting in memory-intensive, computationally slow optimization [19]. PINN approaches become increasingly inefficient in high spatial dimensions and are primarily demonstrated on 1D and 2D problems. PINNs are designed for forward simulation, not optimal control; applying PINNs to control requires a separate RL loop, and the PINN surrogate may not capture dynamics most relevant to control optimization.

Deep Operator Networks (DeepONets) represent a recent advance combining neural networks with operator theory [24]. DeepONets learn mappings between infinite-dimensional function spaces using a branch-trunk architecture where the branch network encodes problem-specific conditioning and the trunk network processes query coordinates. This architecture is grounded in universal approximation theory for operators and provides true operator learning capabilities [25]. Once trained, DeepONet inference can be substantially faster than repeated PDE simulation, supporting rapid evaluations within iterative control workflows [26]. Recent work explores combining DeepONet surrogates with RL by training a DeepONet on precomputed PDE solutions via supervised learning, then using this surrogate as the environment simulator in an RL loop [25].

However, critical limitations persist when combining DeepONet surrogates with RL [18]. Using a neural operator as the RL environment simulator introduces a new layer of artifact exploitation, where policies may overfit to idiosyncrasies of the DeepONet rather than physical principles, failing upon deployment with more accurate simulators or real systems. DeepONets alone do not enforce physics during RL training, allowing agents to discover policies violating conservation laws or exhibiting unrealistic behavior. Neither DeepONet nor standard RL guarantees operational constraint satisfaction, requiring post-hoc constraint checking or penalty methods. DeepONets trained on deterministic trajectories may not capture variability in real deployments, causing RL policies learned on deterministic surrogates to fail under environmental noise [6]. Standard RL optimizes scalar rewards; managing water quality while minimizing costs and ensuring robustness requires

sophisticated reward design, often leading to fragile policies biased toward the highest-weighted objectives.

To address the tension between computational efficiency, physical fidelity and operational feasibility, we present a PI operator assisted dual-critic RL framework. Related developments in PI operator learning and physics guided learning architectures motivate this integration [27,28]. The methodological contribution lies in coupling a one-step PI-DeepONet with separate environmental and physics diagnostic value estimators, validation selected advantage fusion and explicit feasibility projection of distributed treatment actions. The environmental critic learns from closed loop stochastic numerical simulator returns whereas the auxiliary critic receives frozen DeepONet diagnostics derived from local ADR consistency, boundary behavior and predicted non-negativity. Control effort and temporal smoothness remain components of the environmental objective, with physics diagnostics treated as a separate category. The numerical ADR simulator is retained for state transitions and final policy evaluation, while DeepONet provides operator level concentration forecasting and the auxiliary diagnostic signal. The framework is evaluated through comparisons with standard PPO, a single combined critic, Dual-data, Dual-Physics-Informed (Dual-PI) and heuristic control baselines under nominal, out of distribution, delay, execution noise and treatment capacity conditions.

2. Methodology

This study presents an integrated framework combining PI operator learning with dual-critic RL for time sensitive PDE-constrained environmental control. The approach comprises four key stages: (1) validation of a numerical ADR simulator through grid convergence and ensemble testing to establish a stable reference environment, (2) training a one-step PI-DeepONet surrogate via supervised data fitting with time discrete ADR consistency, boundary, and non-negativity regularization to enable rapid concentration field prediction, (3) optimization of a dual-critic PPO agent that leverages environmental returns and frozen surrogate physics diagnostic guidance, and (4) projection of each distributed treatment action onto non-negativity, pointwise actuator, and integral capacity constraints before application to the numerical simulator. The simulator employs upwind differencing for advection, central differencing for diffusion, and explicit Euler-Maruyama integration with spatially correlated Gaussian noise to capture environmental variability. The DeepONet architecture combines supervised one-step learning with physical consistency regularization rather than enforcing the governing equation exactly. At the same time, the dual-critic architecture maintains separate environmental and physics diagnostic value functions, whose standardized advantages are combined using validation selected weights within PPO and GAE. The numerical ADR simulator remains the state-transition and final evaluation environment, while the frozen DeepONet supplies one-step forecasting and the auxiliary diagnostic signal. This unified design supports computationally efficient prediction, explicit action feasibility, and safety oriented evaluation within a controlled stochastic PDE benchmark [9,24,29].

2.1. Simulation implementation

The dynamics of pollutant concentration in a riverine domain are governed by a one dimensional spatiotemporal partial differential equation (PDE) that combines advective transport, molecular diffusion, chemical degradation, pollution source injection and control induced removal [2,3]. The governing equation is expressed as:

$$\frac{\partial c}{\partial t} + v\frac{\partial c}{\partial x} = D\frac{\partial^2 c}{\partial x^2} - \lambda c + f(x,t) - u(x,t) + \xi(x,t) \tag{1}$$

where $c(x,t) : [0,L]\times[0,T] \to \mathbb{R} \geq 0$ denotes the pollutant concentration (mass per unit volume) at spatial position x and time t. The physical parameters are v (velocity of advective flow in m/s), D (molecular diffusion coefficient in m^2/s) and λ (first-order decay rate constant in s^{-1}) [21]. The source term $f(x,t) \geq 0$ represents external pollutant injection from point sources such as industrial outfalls or wastewater treatment discharges (mass per unit volume per unit time), while $u(x,t) \geq 0$ denotes the control action representing pollutant removal through treatment or remediation (mass per unit volume per unit time) [4]. The stochastic forcing term $\xi(x,t)$ accounts for environmental variability and unmodeled disturbances and is represented by spatially correlated Gaussian increments sampled independently at each numerical time step [6].

This equation is supplemented by mixed boundary conditions, a Dirichlet condition at the upstream boundary ($x = 0$) specifying the background inflow concentration $c(0,t) = c_{\text{in}}$ and a zero gradient Neumann condition at the downstream boundary ($x = L$) representing open water outflow with no diffusive flux accumulation [2]. The initial condition $c(x,0) = c_0(x)$ describes the spatial distribution of baseline pollution at $t = 0$. The first order decay term $-\lambda c$ is physically justified by empirical evidence that many aquatic pollutants, including organic contaminants, nutrients and certain pathogens, degrade according to pseudo-first-order kinetics when other reactants are in excess [30,31]. This model formulation provides a parsimonious yet realistic representation of river pollution dynamics suitable for RL based control design. The model serves as a controlled one dimensional synthetic benchmark for distributed treatment allocation. It is designed to provide a general evaluation framework independent of any specific river, hydraulic geometry, treatment facility or regulatory jurisdiction. Accordingly the reported constraints and compliance measures represent benchmark definitions rather than field certification.

2.1.1. Spatial and temporal gridding

A uniform finite-difference grid in space is employed as: $x_i = i\Delta x$ for $i = 0, 1, \ldots, N-1$, where $\Delta x = L/(N-1)$ is the spatial step size and N is the number of grid points. Similarly, time is discretized uniformly: $t^n = n\Delta t$ for $n = 0, 1, \ldots, M$, where Δt is the temporal step size. The discrete concentration field at time step n is denoted $c_i^n = c(x_i, t^n)$.

2.1.2. Advection discretization: Upwind finite differences

The advective term $-v\frac{\partial c}{\partial x}$ is discretized using the upwind (upstream) finite difference scheme, which is crucial for stability when solving hyperbolic conservation laws [5]. The upwind direction depends on the sign of the flow velocity:
for $v > 0$ (flow from left to right):

$$v\frac{\partial c}{\partial x}\bigg|_i^{\text{upwind}} = v\frac{c_i^n - c_{i-1}^n}{\Delta x} \tag{2}$$

for $v < 0$ (flow from right to left):

$$v\frac{\partial c}{\partial x}\bigg|_i^{\text{upwind}} = v\frac{c_{i+1}^n - c_i^n}{\Delta x} \tag{3}$$

This upwind approach is mathematically justified by stability analysis: the upwind scheme is conditionally stable under the standard Courant-Friedrichs-Lewy (CFL) restriction when $|\sigma| \leq 1$, where $\sigma = v\Delta t/\Delta x$ is the Courant number [32]. The CFL condition ensures that the numerical domain of dependence encompasses the true mathematical domain of dependence, preventing the growth of discretization errors [5]. Because the diffusion term is also advanced explicitly, the selected time step must additionally satisfy the corresponding explicit diffusion stability restriction. Truncation error analysis shows that the upwind scheme achieves first order spatial accuracy, $\mathcal{O}(\Delta x)$ [33].

2.1.3. Diffusion discretization: Central differences with ghost cells

The diffusive term $D\frac{\partial^2 c}{\partial x^2}$ is approximated using centered finite differences, which provide second-order spatial accuracy:

$$D\frac{\partial^2 c}{\partial x^2}\bigg|_i^n \approx D\frac{c_{i+1}^n - 2c_i^n + c_{i-1}^n}{(\Delta x)^2} \tag{4}$$

To enforce the boundary conditions, ghost cells are incorporated outside the computational domain. At the upstream boundary ($x = 0$),

we impose a Dirichlet condition using a ghost cell value: $c^n_{-1} = c_{\text{in}}$ (the prescribed inflow concentration) [2,3]. At the downstream boundary ($x = L$), a homogeneous Neumann boundary condition is applied by setting $c^n_N = c^n_{N-1}$, ensuring a zero spatial gradient ($\partial c/\partial x = 0$) and preventing artificial fluxes at the outflow. This mixed boundary condition approach is appropriate for rivers where downstream regions extend beyond the computational domain and no accumulation occurs at the open boundary [3].

2.1.4. Temporal integration: Explicit Euler-Maruyama method

To advance the solution in time, the explicit Euler–Maruyama scheme is employed, a standard first-order accurate method for solving stochastic differential equations [7,34]. The update rule at each time step is:

$$c_i^{n+1} = c_i^n + \Delta t\left[-v\frac{\partial c}{\partial x}\bigg|_i^n + D\frac{\partial^2 c}{\partial x^2}\bigg|_i^n - \lambda c_i^n + f_i^n - u_i^n\right] + \delta W_i^n \tag{5}$$

where δW_i^n represents the stochastic increment at spatial location i and time step n. The scheme is first order accurate in time for the deterministic part (truncation error $\mathcal{O}(\Delta t)$) and generally has strong convergence order 1/2 for stochastic differential equations, although higher convergence rates are possible for additive noise under suitable regularity assumptions [7].

2.1.5. Stochastic noise generation: Spatially correlated Gaussian increments

To capture realistic environmental variability, the noise term $\xi(x,t)$ is modeled as a spatially correlated Gaussian process. The discrete stochastic increments δW_i^n are sampled from a multivariate normal distribution whose covariance structure is defined by a kernel function $K(x_i, x_j)$ [35]. Three spatial covariance kernel options are implemented:

(i). **Exponential kernel:** $K(x_i, x_j) = \sigma^2 \exp(-|x_i - x_j|/\ell)$

(ii). **Matérn 3/2 kernel:** $K(x_i, x_j) = \sigma^2(1 + \sqrt{3}|x_i - x_j|/\ell)\exp(-\sqrt{3}|x_i - x_j|/\ell)$

(iii). **Matérn 5/2 kernel:** $K(x_i, x_j) = \sigma^2(1 + \sqrt{5}|x_i - x_j|/\ell + 5|x_i - x_j|^2/(3\ell^2))\exp(-\sqrt{5}|x_i - x_j|/\ell)$

where σ is the noise standard deviation and ℓ is the correlation length scale [36]. The Matérn kernels are preferred because they allow independent control of smoothness (via the parameter ν, taking values 3/2 or 5/2) and correlation length, providing greater flexibility than the exponential kernel [29].

To generate samples from the correlated Gaussian process, the Cholesky decomposition L of the covariance matrix K is computed, and the stochastic increment is then formed as:

$$\delta \mathbf{W}^n = \sqrt{\Delta t}\, L\, \mathbf{z}^n \tag{6}$$

where $\mathbf{z}^n$ is a vector of independent standard normal random variables. If the covariance matrix is nearly singular (which can occur for large correlation lengths relative to the spatial domain or grid spacing), a small regularization term ϵI (where $\epsilon \sim 10^{-12}$) is added to ensure numerical stability.

2.1.6. Initial condition generation

The initial spatial profile of pollution $c_0(x)$ is generated using Gaussian process sampling to produce spatially correlated initial fields that reflect realistic background contamination patterns. Rather than prescribing uniform or delta function-like initial concentrations, this approach generates smooth, spatially heterogeneous profiles that are physically plausible. The initial condition is sampled as:

$$c_0(x) = \mu_c + L\mathbf{z} \tag{7}$$

where μ_c is the mean concentration, L is the Cholesky factor of a spatial kernel matrix K, and $\mathbf{z}$ is a vector of independent standard normal random variables. The choice of kernel function (exponential, Matérn 3/2, or Matérn 5/2) and correlation length ℓ controls the spatial smoothness of the initial condition. This sampling procedure ensures that the initial condition space is sufficiently diverse to train robust control policies that generalize across varying baseline pollution profiles.

2.1.7. Pollution source specification

The source term $f(x,t) = S(x)g(t)$ is modeled as a product of spatial and temporal components, enabling representation of multiple outfalls with time-varying discharge rates. The spatial footprint $S(x)$ is constructed as a superposition of Gaussian functions:

$$S(x) = \sum_{k=1}^{K} \alpha_k \exp\left(-\frac{1}{2}\left(\frac{x - x_k}{\sigma_k}\right)^2\right) \tag{8}$$

where K is the number of source locations, x_k is the position of the kth source, σ_k is its spatial width, and α_k is its strength factor. The Gaussian form allows a smooth representation of spatially distributed outfalls, which is more realistic than point sources.

The temporal modulation $g(t)$ captures multiple timescale variations in pollution loading:

$$g(t) = g_{\text{base}}\left[1 + A_{\text{diurn}}\sin(2\pi t/P_{\text{diurn}}) + A_{\text{seas}}\sin(2\pi t/P_{\text{seas}})\right] + g_{\text{pulse}}(t) \tag{9}$$

where g_{base} is the baseline discharge level, A_{diurn} and P_{diurn} are the diurnal oscillation amplitude and period (typically 24 h), A_{seas} and P_{seas} are seasonal modulation parameters, and $g_{\text{pulse}}(t)$ represents episodic storm events or process upsets.

The episodic pulses are generated as a Poisson point process: event arrival times are drawn from an exponential distribution with mean interval τ_{event}, and event amplitudes are drawn from a normal distribution with mean μ_{event} and standard deviation σ_{event}. This multi-timescale source model creates realistic pollution scenarios that challenge control policies across varying environmental conditions. Within the 10-s benchmark horizon, the periodic components represent short-period and long-period variations in normalized simulator time with daily and seasonal cycles serving as conceptual field scale analogies.

2.2. Dataset generation and scenario-level separation

A total of 250 in-distribution scenarios was generated on the same 10-s temporal horizon and 200 points spatial grid used for the simulator analysis. Latin hypercube sampling was used to obtain space filling coverage of the transport parameters, source configurations, initial conditions, stochastic forcing and environmental effort coefficients [37]. The in-distribution transport ranges were

$$v \in [0.05, 0.20]\ \text{m s}^{-1},\ D \in [0.005, 0.020]\ \text{m}^2\,\text{s}^{-1},\ \lambda \in [0.01, 0.08]\ \text{s}^{-1} \tag{10}$$

Scenario identity was used as the unit of data separation rather than individual temporal slices. A fixed manifest assigned 200 scenarios to training, 25 to validation and 25 to an untouched in-distribution test set. All temporal transitions from a given scenario remained within the same split, ensuring independence between trajectories generated under identical physical and source conditions. Deterministic one-step transitions were used as supervised DeepONet targets, whereas stochastic numerical simulator rollouts were used for controller training and evaluation.

A separate set of 20 independently generated OOD scenarios was used for held out evaluation. It contained five high velocity, five high diffusion, five high reaction and five combined high cases outside the corresponding training ranges. Dataset size selection, surrogate selection, critic weight selection, and policy checkpoint selection used only the training and validation scenarios. The in-distribution test and OOD scenarios were used only for held-out evaluation.

Dataset size sensitivity was evaluated using 20, 100 and 200 training scenarios with the same validation set. The 200 scenarios configuration was selected from validation performance. An independent 20 scenarios deterministic/stochastic subset was used exclusively for numerical simulator validation. Table 1 summarizes the scenario level data partition

Table 1
Scenario-level data separation and experimental role.

Dataset	Total	Train	Validation	Test	Purpose
In-distribution	250	200	25	25	Training, selection, untouched evaluation
OOD	20	–	–	20	External generalization evaluation
Simulator-validation subset	20	–	–	–	Grid and stochastic checks

Table 2
Parameter values are sampled via Latin hypercube or uniform distributions as appropriate.

Parameter	Range	Unit
Flow velocity, v	[0.05, 0.20]	$\mathrm{m\,s^{-1}}$
Diffusion coefficient, D	[0.005, 0.020]	$\mathrm{m^2\,s^{-1}}$
Decay rate, λ	[0.01, 0.08]	$\mathrm{s^{-1}}$
Environmental effort coefficient, β_{lin}	[0.1, 2.0]	Dimensionless

and the role of each dataset, while Table 2 lists the sampled parameter ranges used for scenario generation.

2.3. DeepONet architecture for one-step temporal encoding

The surrogate uses a branch-trunk architecture formulated as a one-step transition operator. For a transition from output time t_k to t_{k+1}, the branch input is

$$\mathbf{u}_{\mathrm{br},k} = \left[\mathbf{x}^{\mathrm{sub}}, \mathbf{c}_0^{\mathrm{sub}}, \mathbf{c}_k^{\mathrm{sub}}, \mathbf{f}_k^{\mathrm{sub}}, \mathbf{u}_k^{\mathrm{sub}}, v, D, \lambda, t_k, \Delta t_k\right]^{\mathsf{T}} \in \mathbb{R}^{165} \tag{11}$$

where each spatial field is subsampled at 32 uniformly distributed locations and $\Delta t_k = t_{k+1} - t_k$. The environmental effort coefficient is not a physical ADR parameter and is therefore excluded from the DeepONet branch vector. The trunk network receives the spatial query and next output time (x, t_{k+1}) and the supervised target is $c(x, t_{k+1})$ on the full 200 points grid.

The factorized prediction remains

$$\widehat{c}(x, t_{k+1} \mid \mathbf{u}_{\mathrm{br},k}) = \sum_{p=1}^{P} b_p(\mathbf{u}_{\mathrm{br},k})\tau_p(x, t_{k+1}), \qquad P = 32 \tag{12}$$

The branch network uses hidden widths 256 and 128, the trunk network uses hidden widths 128 and 64, both use Gaussian Error Linear Units (GELU) activation, and the common latent width is 32 [24]. Full field trunk queries are evaluated in chunks of at most 4096 points to limit GPU memory consumption.

2.4. Composite physics-informed loss objective

The supervised full-grid one-step loss is

$$\mathcal{L}_{\mathrm{data}} = \frac{1}{BN_x} \sum_{b=1}^{B} \sum_{i=1}^{N_x} \left[\widehat{c}_b(x_i, t_{k+1}) - c_b^{\mathrm{true}}(x_i, t_{k+1})\right]^2 \tag{13}$$

To evaluate time discrete ADR consistency, define

$$\mathcal{L}_{\mathrm{ADR}}(c) = v\frac{\partial c}{\partial x} - D\frac{\partial^2 c}{\partial x^2} + \lambda c \tag{14}$$

Using $\theta = 0.5$, the output step residual is

$$\mathcal{R}_k(c_{k+1}) = \frac{c_{k+1} - c_k}{\Delta t_k} + \theta\mathcal{L}_{\mathrm{ADR}}(c_{k+1}) + (1-\theta)\mathcal{L}_{\mathrm{ADR}}(c_k) - \left[\theta f_{k+1} + (1-\theta) f_k\right] + \left[\theta u_{k+1} + (1-\theta) u_k\right] \tag{15}$$

Because the numerical target can have a non-zero value of Eq. (15) under output grid differentiation, the surrogate added defect is

$$D_k = \mathcal{R}_k(\widehat{c}_{k+1}) - \mathcal{R}_k(c_{k+1}^{\mathrm{true}}) \tag{16}$$

and

$$\mathcal{L}_{\mathrm{def}} = \frac{1}{BN_{\mathrm{col}}} \sum_{b=1}^{B} \sum_{j=1}^{N_{\mathrm{col}}} D_{b,k}(x_j)^2 \tag{17}$$

Boundary and non-negativity terms are

$$\mathcal{L}_{\mathrm{bc}} = \frac{1}{B} \sum_{b=1}^{B} \left[\left|\widehat{c}_b(0, t_{k+1}) - c_{\mathrm{in}}\right|^2 + \left|\frac{\partial \widehat{c}_b}{\partial x}(L, t_{k+1})\right|^2\right] \tag{18}$$

$$\mathcal{L}_{+} = \frac{1}{BN_x} \sum_{b=1}^{B} \sum_{i=1}^{N_x} \mathrm{ReLU}\left[-\widehat{c}_b(x_i, t_{k+1})\right]^2 \tag{19}$$

The PI continuation minimizes

$$\mathcal{L}_{\mathrm{PI}} = \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{def}}\mathcal{L}_{\mathrm{def}} + \lambda_{\mathrm{bc}}\mathcal{L}_{\mathrm{bc}} + \lambda_{+}\mathcal{L}_{+} \tag{20}$$

The defect in Eq. (16) is a supervised physical consistency diagnostic relative to the numerical target. It is not a label-free continuous PDE residual and does not imply exact enforcement of the governing equation.

2.5. Training protocol and optimization dynamics

A data only DeepONet was first trained on the fixed 200 scenarios training split, while the 25 validation scenarios were used exclusively for checkpoint selection and hyperparameter evaluation. Dataset size experiments compared 20, 100 and 200 training scenarios using the same scenario separated validation protocol. Dataset size and physics schedule experiments were repeated using random seeds 11, 22 and 33.

Optimization utilizes the adaptive moment estimation algorithm with decoupled weight decay (AdamW) regularization [38]:

$$\theta_{t+1} = \theta_t - \alpha\left(\frac{\mathbf{m}_t}{\sqrt{\mathbf{v}_t} + \epsilon}\right) - \lambda_{\mathrm{wd}}\theta_t \tag{21}$$

For the initial data only training stage, the AdamW configuration used learning rate $\alpha = 10^{-3}$, weight-decay coefficient $\lambda_{\mathrm{wd}} = 10^{-4}$, momentum parameters $(\beta_1, \beta_2) = (0.9, 0.999)$ and numerical constant $\epsilon = 10^{-8}$. Gradient clipping with maximum norm $\|\nabla\mathcal{L}\|_2 \le 1.0$ was applied to stabilize optimization.

The physics schedule study compared data only training with physics-diagnostic evaluation every 1, 2, 4, 8 or 16 batches. For each schedule, validation relative L_2 error, defect corrected ADR with Root Mean Square Error (ADR-RMSE), boundary consistency, negative-prediction fraction and epoch time were recorded. The every eight-batch schedule provided the selected validation compromise between predictive accuracy, physical consistency diagnostics, and computational cost. This schedule was fixed before the untouched test and OOD datasets were evaluated.

The selected PI model was warm started from the frozen data only checkpoint. The selected continuation used a maximum defect corrected ADR weight of 0.01, boundary weight 1.0, non-negativity weight 0.1, and 128 physics points per sample. The learning rate decreased from 8×10^{-6} to 10^{-6}. To prevent the physics component from overwhelming the supervised objective, the norm of the physics gradient was restricted to at most 0.03 times the data gradient norm. Conflicting data and physics gradients were handled using projected conflicting gradients [39]. Validation only Pareto selection identified epoch 29 as the final PI checkpoint.

Checkpoint acceptance was governed by the following prespecified validation criterion:

$$\frac{E_{L_2}^{\mathrm{PI}}}{E_{L_2}^{\mathrm{data}}} \le 1.15, \qquad \frac{E_{\mathrm{def}}^{\mathrm{PI}}}{E_{\mathrm{def}}^{\mathrm{data}}} \le 0.80 \tag{22}$$

The first condition protects pointwise predictive accuracy, whereas the second represents the prespecified 20% reduction in defect corrected ADR error. Checkpoint selection used validation data only after which the selected data only and PI models were evaluated on the 25 scenarios in-distribution test set and the independent 20 scenarios OOD set. The complete training, validation, physics-schedule selection, and checkpoint-selection procedure for the one-step PI-DeepONet is summarized in Algorithm 1.

Algorithm 1 Training procedure for the scenario-separated one-step DeepONet surrogate.

Input: $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{val}}$; fixed scenario manifests, seeds $\{11, 22, 33\}$, and physics schedules $S \in \{1, 2, 4, 8, 16\}$
Train the data only DeepONet using $\mathcal{L}_{\text{data}}$
Select and freeze the data only checkpoint using validation data only
for seed $q \in \{11, 22, 33\}$ **do**
  for physics schedule $S \in \{1, 2, 4, 8, 16\}$ **do**
    Initialize the model from the same frozen data only checkpoint
    for epoch = 1 to 100 **do**
      for batch in DataLoader($\mathcal{D}_{\text{train}}$, B) **do**
        Branch: $\mathbf{b} = \text{Branch}(\mathbf{u}_{\text{br}}) \in \mathbb{R}^{32}$
        Trunk: $\boldsymbol{\tau} = \text{Trunk}(x, t_{k+1}) \in \mathbb{R}^{32}$ (chunked, chunk size 4096)
        Prediction: $\widehat{\mathbf{c}}_{k+1} = \sum_p b_p \boldsymbol{\tau}_p \in \mathbb{R}^{200}$
        Data loss: $\mathcal{L}_{\text{data}} = \dfrac{1}{BN_x} \sum_{b,i} \left[\widehat{c}_{b,k+1}(x_i) - c^{\text{true}}_{b,k+1}(x_i) \right]^2$
        if (global_step mod S) $= 0$ **then**
          Compute the target-relative defect-corrected ADR loss $\mathcal{L}_{\text{def}}$
          Compute the boundary-consistency loss $\mathcal{L}_{\text{bc}}$
          Restrict the physics gradient norm to at most 0.03 of the data gradient norm
          Apply projected conflicting gradients to conflicting data and physics gradients
        else
          $\mathcal{L}_{\text{def}} = 0$ and $\mathcal{L}_{\text{bc}} = 0$
        end if
        Compute the non-negativity loss $\mathcal{L}_{+}$
        Total loss: $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{def}}\mathcal{L}_{\text{def}} + \lambda_{\text{bc}}\mathcal{L}_{\text{bc}} + \lambda_{+}\mathcal{L}_{+}$
        Backward: $\mathcal{L}$.backward(); apply gradient control; Opt.step()
      end for
      Validation: compute relative L_2 error, defect-corrected ADR-RMSE, boundary errors, negative-prediction fraction, and epoch time
      if the validation selection criterion improves **then**
        Save checkpoint
      end if
    end for
    Store the validation results for seed q and schedule S
  end for
end for
Select the every eight batch schedule using validation results only
Warm start the final physics informed continuation from the frozen data only checkpoint
Train using $\lambda_{\text{def}} \le 0.01$, $\lambda_{\text{bc}} = 1.0$, $\lambda_{+} = 0.1$, 128 physics points per sample and learning rate $8 \times 10^{-6} \rightarrow 10^{-6}$
Select epoch 29 using validation only Pareto selection
Apply the prespecified criterion in Eq. (22)
Output: frozen data only and PI checkpoints for untouched test and OOD evaluation

2.6. Reinforcement learning framework

The river pollution control policy is trained via PPO with a dual-critic architecture that combines environmental returns from the numerical ADR simulator with an auxiliary physics diagnostic signal obtained from the frozen DeepONet [9]. The numerical ADR simulator remains the state transition environment and the final controller evaluation model. The frozen DeepONet is not used to replace the simulator; instead, it provides one-step concentration forecasting and a surrogate derived local ADR consistency diagnostic. The agent operates in a continuous action space via diagonal Gaussian policies with state observations normalized online using Welford's algorithm to improve numerical stability and accelerate convergence. Advantages are estimated via Generalized Advantage Estimation (GAE) with bias-variance balancing ($\lambda_{\text{GAE}} = 0.95, \gamma = 0.99$), and the policy is regularized through an entropy bonus ($\beta_{\text{ent}} = 0.01$) to discourage premature loss of exploration during training [15,40].

The use of diagonal Gaussian policies in continuous action space is particularly well-suited for environmental control applications, where distributed treatment amplitudes vary continuously across the spatial domain. The sampled actor coefficients are decoded through fixed Gaussian Radial Basis Functions (RBFs) to generate spatially smooth treatment fields. The entropy term supports exploration during optimization but is not interpreted as guaranteeing policy robustness or OOD generalization. The dual-critic architecture represents a key departure from standard single-critic RL: rather than relying on a single value function, this work maintains separate environmental and physics-diagnostic value estimates. The environmental critic learns from monitored concentration exceedance, treatment effort, and temporal smoothness evaluated by the numerical simulator, whereas the auxiliary critic learns from the frozen-surrogate diagnostic combining local ADR consistency, boundary behavior, and predicted non-negativity. Treatment cost and action smoothness are therefore components of the environmental objective and are not classified as physics diagnostics.

The two critics are not combined by directly averaging their value predictions. Instead, environmental and physics-diagnostic generalized advantages are computed and standardized independently, after which they are combined as

$$\widetilde{A}_t = w_{\text{env}} \widetilde{A}_t^{(\text{env})} + w_{\text{phys}} \widetilde{A}_t^{(\text{phys})} \tag{23}$$

where five candidate weight pairs were compared using validation scenarios only. The selected values, $w_{\text{env}} = 0.25$ and $w_{\text{phys}} = 0.75$, were frozen before test and OOD evaluation. These weights are treated as an empirical selection for the present benchmark rather than as universally optimal critic weights.

The physics diagnostic critic provides an auxiliary learning signal and does not impose a hard PDE-constraint on the policy. Operational feasibility is enforced separately by projecting every decoded treatment field onto non-negativity, pointwise actuator, and integral capacity constraints before the action is applied to the numerical simulator.

2.6.1. Online observation normalization and policy parameterization

The observation contains 100 uniformly sampled concentration values, normalized episode time and the three ADR parameters:

$$\mathbf{s}_t = \left[c(x_1, t), \ldots, c(x_{100}, t), t/T, v, D, \lambda \right]^{\top} \in \mathbb{R}^{104} \tag{24}$$

State observations $\mathbf{s}_t \in \mathbb{R}^{104}$ are normalized during episodes via incremental Welford's algorithm [41], which computes running mean and variance without storing all historical observations:

$$\boldsymbol{\mu}_n = \boldsymbol{\mu}_{n-1} + \frac{\mathbf{s}_n - \boldsymbol{\mu}_{n-1}}{n}, \qquad \bar{\mathbf{s}}_t = \frac{\mathbf{s}_t - \boldsymbol{\mu}}{\sqrt{\boldsymbol{\sigma}^2 + 10^{-6}}} \tag{25}$$

Online normalization is used to reduce differences in scale among the concentration observations, normalized time, and ADR parameters, thereby improving the numerical conditioning of the actor and critic

inputs. Welford's algorithm is memory efficient and numerically stable for incrementally collected trajectories. The small regularization constant 10^{-6} prevents division by zero when an observation component has very small running variance. The running mean, variance and observation count are stored with each selected policy checkpoint and are frozen during test, OOD and operational stress evaluation.

The policy $\pi_\theta(\mathbf{a} \mid \bar{\mathbf{s}}_t)$ is parameterized as a diagonal Gaussian with actor network outputs $\boldsymbol{\mu}_\theta(\bar{\mathbf{s}}_t)$ and $\log \boldsymbol{\sigma}_\theta(\bar{\mathbf{s}}_t)$ representing the mean and log standard deviation of each action dimension. The actor samples 16 latent treatment amplitudes, which are decoded through fixed Gaussian RBFs with centers x_j and width $\sigma_b = 0.05L$:

$$u_{\mathrm{raw}}(x) = \sum_{j=1}^{16} a_j \phi_j(x; x_j, 0.05L), \qquad \mathbf{a} \sim \pi_\theta\left(\cdot \mid \bar{\mathbf{s}}_t\right) \tag{26}$$

The RBF parameterization provides a smooth, low dimensional representation of the distributed treatment field [42]. If the RL agent directly produced independent pointwise values $u(x_i)$ at every spatial location, the resulting treatment field could exhibit rapid spatial variations or discontinuities. By representing the action as a linear combination of smooth Gaussian basis functions, the policy can express upstream, downstream or multimodal treatment allocations using only 16 latent coefficients rather than 200 independent pointwise commands. The fixed centers and width reduce the action space dimension and avoid introducing additional trainable basis parameters. However RBF decoding alone does not guarantee operational feasibility. The decoded field $u_{\mathrm{raw}}(x)$ is therefore treated as an unconstrained command and is subsequently projected onto non-negativity, pointwise actuator and integral capacity constraints before being applied to the numerical ADR simulator.

2.6.2. Action feasibility and operational constraints

Every decoded command is projected before application to the numerical simulator. First, pointwise clipping gives

$$\widetilde{u}_t(x) = \max\left[0, \min\left(u_{\max}, u_{\mathrm{raw},t}(x)\right)\right] \tag{27}$$

The pointwise clipped field is then projected onto the integral capacity constraint:

$$u_t(x) = \widetilde{u}_t(x) \min\left[1, \frac{U_{\mathrm{cap}}}{\int_0^L \widetilde{u}_t(s)\, ds + \varepsilon}\right] \tag{28}$$

The nominal limits are

$$0 \le u_t(x) \le u_{\max} = 2.0, \qquad \int_0^L u_t(x)\, dx \le U_{\mathrm{cap}} = 0.75 \tag{29}$$

For operational-stress evaluation, response delay or multiplicative execution noise is applied to the commanded treatment field, after which the projection in Eq. (28) is reapplied. Therefore, the treatment finally supplied to the numerical simulator satisfies non-negativity, the pointwise model limit and the integral capacity limit under every evaluated condition. These are simulator level feasibility constraints and are not presented as field-certified actuator specifications. The comparison includes classical heuristic controls and multiple learned ablations.

2.6.3. Environmental and physics diagnostic critics

The environmental and physics diagnostic critics are trained from separate reward streams. The monitoring set and benchmark concentration limit are

$$\mathcal{M} = \{0.8L, 0.9L\}, \qquad c_{\mathrm{lim}} = 1.0\ \mathrm{mg\,L^{-1}} \tag{30}$$

For one control interval $[t, t + \Delta t_c]$, monitored exceedance, linear treatment effort, quadratic effort, and temporal treatment variation are defined as

$$E_t = \int_t^{t+\Delta t_c} \frac{1}{|\mathcal{M}|} \sum_{x_m \in \mathcal{M}} \left[c(x_m, \tau) - c_{\mathrm{lim}}\right]_+ d\tau \tag{31}$$

$$J_t^{\mathrm{lin}} = \int_t^{t+\Delta t_c} \int_0^L u(x, \tau)\, dx\, d\tau \tag{32}$$

$$J_t^{\mathrm{quad}} = \int_t^{t+\Delta t_c} \int_0^L u(x, \tau)^2\, dx\, d\tau \tag{33}$$

$$J_t^{\mathrm{smooth}} = \Delta t_c \int_0^L \left[u_t(x) - u_{t-1}(x)\right]^2 dx \tag{34}$$

The environmental reward is

$$r_t^{\mathrm{env}} = -\left(E_t + \beta_{\mathrm{lin}} J_t^{\mathrm{lin}} + 0.01 J_t^{\mathrm{quad}} + 0.005 J_t^{\mathrm{smooth}}\right) \tag{35}$$

where β_{lin} is the scenario dependent linear treatment cost coefficient used by the environmental objective. Treatment effort, quadratic action regularization, and temporal smoothness are environmental objectives and are not labeled as physics. Before each numerical transition the frozen DeepONet is evaluated at 32 selected spatial points using the current branch state and the finally projected treatment command.

The local ADR consistency proxy is

$$\mathcal{R}_{\mathrm{proxy}} = \frac{\partial \hat{c}}{\partial t_{\mathrm{trunk}}} + v \frac{\partial \hat{c}}{\partial x} - D \frac{\partial^2 \hat{c}}{\partial x^2} + \lambda \hat{c} - f + u \tag{36}$$

The corresponding surrogate diagnostic and reward are

$$\Phi_t = \frac{\mathrm{RMSE}\left(\mathcal{R}_{\mathrm{proxy}}\right)}{0.4} + 0.25 E_{\mathrm{bc},t} + 0.25 E_{-,t} \tag{37}$$

$$r_t^{\mathrm{phys}} = -\Delta t_c \Phi_t \tag{38}$$

where $E_{\mathrm{bc},t}$ measures the upstream Dirichlet and downstream gradient mismatch and $E_{-,t}$ measures the predicted negative concentration. The branch state is held fixed while the trunk time coordinate is differentiated in Eq. (36). Consequently, this auxiliary signal is described as a frozen surrogate local ADR consistency diagnostic.

2.6.4. Generalized advantage estimation and validation selected critic fusion

Separate GAE sequences are computed for the environmental and physics diagnostic critics [15]. For $j \in \{\mathrm{env}, \mathrm{phys}\}$, the temporal difference residual is

$$\delta_t^{(j)} = r_t^{(j)} + \gamma V_j\left(\bar{\mathbf{s}}_{t+1}\right) - V_j\left(\bar{\mathbf{s}}_t\right) \tag{39}$$

and the finite horizon GAE sequence is

$$\widehat{A}_t^{(j)} = \sum_{\ell=0}^{H-t-1} \left(\gamma \lambda_{\mathrm{GAE}}\right)^\ell \delta_{t+\ell}^{(j)} \tag{40}$$

Each advantage stream is standardized independently:

$$\widetilde{A}_t^{(j)} = \frac{\widehat{A}_t^{(j)} - \mu_j}{\sigma_j + \varepsilon} \tag{41}$$

The actor uses the weighted advantage

$$\widetilde{A}_t = w_{\mathrm{env}} \widetilde{A}_t^{(\mathrm{env})} + w_{\mathrm{phys}} \widetilde{A}_t^{(\mathrm{phys})}, \qquad w_{\mathrm{env}} + w_{\mathrm{phys}} = 1 \tag{42}$$

Five candidate pairs,

$$(1, 0), \quad (0.75, 0.25), \quad (0.5, 0.5), \quad (0.25, 0.75), \quad (0, 1) \tag{43}$$

were screened using validation scenarios only. The critic weight screening was conducted using single specific seed, after which the selected pair was used in the three seed principal controller experiments.

The selected pair minimized the normalized distance to the joint environmental and diagnostic validation ideal point:

$$w_{\mathrm{env}} = 0.25, \qquad w_{\mathrm{phys}} = 0.75 \tag{44}$$

The selected weights were frozen before the in-distribution test, OOD and operational stress evaluations. They are an empirical selection for the present benchmark rather than universally optimal critic weights.

Policy updates employ PPO with clipped probability ratios [9]:

$$\mathcal{L}^{\text{clip}} = \mathbb{E}_t\left[\min\left(r_t(\theta)\widetilde{A}_t, \text{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\widetilde{A}_t\right)\right] \tag{45}$$

where

$$r_t(\theta) = \frac{\pi_\theta\left(\mathbf{a}_t \mid \bar{\mathbf{s}}_t\right)}{\pi_{\theta_{\text{old}}}\left(\mathbf{a}_t \mid \bar{\mathbf{s}}_t\right)}, \qquad \epsilon = 0.2 \tag{46}$$

The clipping operation limits the change in the policy probability ratio to the interval $[0.8, 1.2]$ when this clipping produces the more conservative objective. This reduces the risk of excessively large policy updates based on high variance or stale advantage estimates.

2.6.5. Complete actor critic objective

The training objective combines policy gradient, dual value losses and entropy regularization:

$$\mathcal{L}_{\text{PPO}} = -\mathcal{L}^{\text{clip}} + 0.5\left(\mathcal{L}_v^{\text{env}} + \mathcal{L}_v^{\text{phys}}\right) - 0.01\mathcal{H}\left[\pi_\theta\right] \tag{47}$$

The environmental and physics diagnostic value losses minimize mean squared error to their respective return targets. The actor and both critics use a learning rate of 3×10^{-4}. PPO uses $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, clipping parameter $\epsilon = 0.2$, entropy coefficient 0.01, target KL divergence 0.02, gradient-norm clipping at 0.5, 10 optimization epochs per update and minibatches of 512 transitions. Each training epoch collects 4096 transitions.

The principal learned controllers were trained for 100 epochs using independent seeds 11, 22 and 33. Validation scenarios were used for policy checkpoint selection. The observation normalization mean, variance and count were stored with each selected checkpoint and frozen during in-distribution test, OOD and operational stress evaluation. Algorithm 2 summarizes the constrained dual-critic PPO training and validation procedure.

Algorithm 2 Constrained dual-critic PPO training.

Input:
  frozen DeepONet, numerical ADR simulator, training and validation scenario manifests and selected critic weights
for seed $q \in \{11, 22, 33\}$ **do**
  Initialize the actor, environmental critic, physics-diagnostic critic and observation normalizer
  for epoch = 1 to 100 **do**
    Load a training scenario and reset the numerical environment
    $buf \leftarrow \{\}$; $steps \leftarrow 0$
    while $steps < 4096$ **do**
      Normalize observation: $\bar{\mathbf{s}}_t = (\mathbf{s}_t - \boldsymbol{\mu})/\sqrt{\boldsymbol{\sigma}^2 + 10^{-6}}$
      Policy sample: $(\mathbf{a}_t, \log \pi_t) = \text{Actor}(\bar{\mathbf{s}}_t)$
      Decode raw treatment: $u_{\text{raw},t} = \text{RBFDecoder}(\mathbf{a}_t)$
      Project the decoded treatment using Eq. (28)
      Evaluate the frozen-surrogate diagnostic and compute r_t^{phys}
      Apply the projected treatment to the numerical ADR simulator and obtain $\mathbf{s}_{t+1}$
      Compute environmental reward r_t^{env}
      Store observation, action, rewards, values, log probability, next observation, and termination flag
      Update the observation normalizer using training data
      $steps \leftarrow steps + 1$
    end while
    Compute separate environmental and physics diagnostic GAE sequences
    Standardize both advantage streams independently
    Fuse the standardized advantages using Eq. (42)
    Update the actor and both critics using clipped PPO
    Evaluate the policy on validation scenarios and save the selected checkpoint and normalization statistics
  end for
end for
Output: three frozen policy checkpoints for untouched test, OOD, and operational-stress evaluation

2.7. Controller baselines and ablations

The closed loop evaluation considered eight controller configurations: no control, proportional control based on monitored exceedance, uniform full capacity treatment, standard PPO with a single environmental critic, single combined PPO trained using a scalar combination of environmental and physics-diagnostic rewards, Dual data using the frozen data only DeepONet, Dual-PI using the selected PI-DeepONet and a physics only single critic controller.

Standard PPO, single combined PPO, Dual-data and Dual-PI were trained independently using random three seeds. The physics only controller is included as supporting evidence because only one independently trained seed was available. The no-control, proportional control and uniform-treatment baselines do not require policy training. Collectively, these comparisons isolate the effects of learned control, critic separation, surrogate assisted decision making and continuation from a PI surrogate.

2.8. Evaluation conditions and parameter regimes

The frozen controllers were evaluated on the 25 untouched in-distribution test scenarios and the 20 independent OOD scenarios. Operational robustness was assessed under five predefined conditions: nominal treatment capacity with $U_{\text{cap}} = 0.75$, a response delay of three control decisions, 5% multiplicative execution noise, reduced treatment capacity with $U_{\text{cap}} = 0.50$ and a combined stress condition incorporating response delay, execution noise and reduced capacity. Under each condition the final control command was projected onto the admissible action space after all disturbances had been applied, while the originally trained policies were retained unchanged throughout the stress test evaluations.

For the parameter regime analysis the in-distribution test scenarios were assigned to fixed low, medium and high value bins according to the velocity v, diffusion coefficient D and reaction rate parameter λ. The independent OOD scenarios were categorized into high velocity, high diffusion, high reaction rate and combined high groups. All parameter bin boundaries and OOD group definitions were established before the controller comparisons and remained fixed throughout the evaluation, independently of the observed policy outcomes.

2.9. Performance metrics and statistical analysis

All water quality metrics use the monitoring set $\mathcal{M} = \{0.8L, 0.9L\}$ and the single benchmark limit $c_{\text{lim}} = 1.0\ \text{mg L}^{-1}$.

For policy training seed q, scenario s, stochastic replicate r, output time t_n and monitoring station m, point-time compliance is

$$C_{\text{point}}^{(q)} = \frac{1}{N_s N_r N_t N_m}\sum_{s=1}^{N_s}\sum_{r=1}^{N_r}\sum_{n=1}^{N_t}\sum_{m=1}^{N_m}\mathbb{I}\left[c_{q,s,r}(x_m, t_n) \le c_{\text{lim}}\right] \tag{48}$$

The reported mean across the $N_q = 3$ independently trained policies is

$$\overline{C}_{\text{point}} = \frac{1}{N_q}\sum_{q=1}^{N_q} C_{\text{point}}^{(q)} \tag{49}$$

Simultaneous all station compliance for policy seed q is

$$C_{\text{sim}}^{(q)} = \frac{1}{N_s N_r N_t}\sum_{s=1}^{N_s}\sum_{r=1}^{N_r}\sum_{n=1}^{N_t}\mathbb{I}\left[\max_m c_{q,s,r}(x_m, t_n) \le c_{\text{lim}}\right] \tag{50}$$

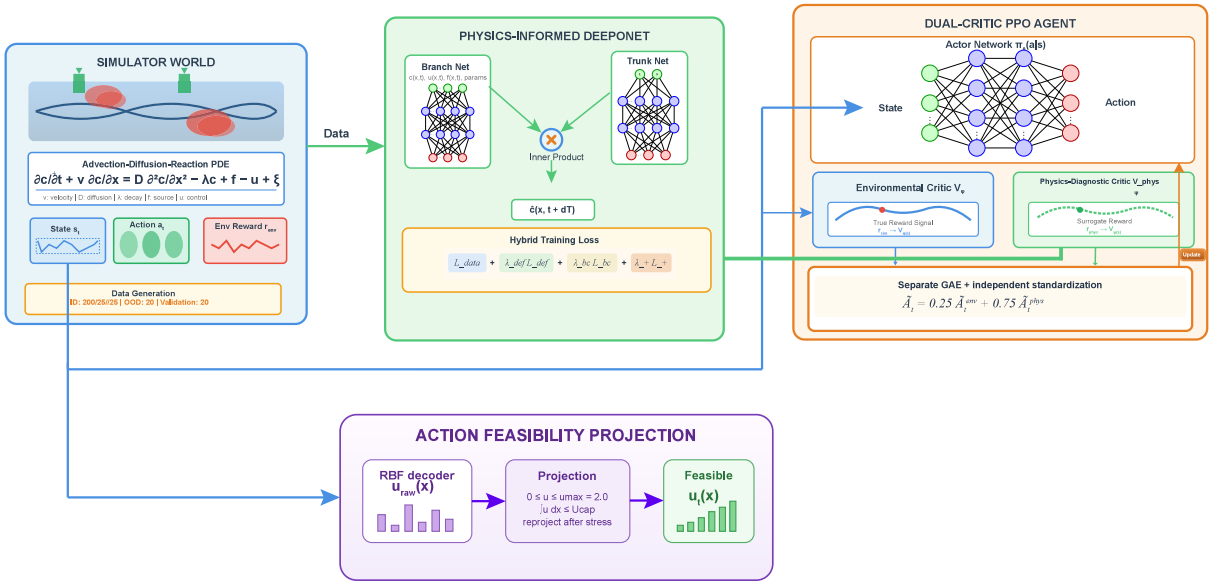


Fig. 1. Integrated ADR-DeepONet dual-critic PPO framework with physics guided learning and feasibility projected treatment actions.

and strict episode compliance is

$$C_{\text{episode}}^{(q)} = \frac{1}{N_s N_r} \sum_{s=1}^{N_s} \sum_{r=1}^{N_r} \mathbb{I}\left[\max_{n,m} c_{q,s,r}(x_m, t_n) \le c_{\text{lim}}\right] \tag{51}$$

All results were evaluated using point-time compliance, denoted by C_{point}. For each closed loop rollout, integrated exceedance, violation duration, peak monitored concentration and treatment effort are

$$E_{\text{int}} = \int_0^T \frac{1}{N_m} \sum_{m=1}^{N_m} \left[c(x_m, t) - c_{\text{lim}}\right]_+ dt \tag{52}$$

$$T_{\text{viol}} = \int_0^T \mathbb{I}\left[\max_m c(x_m, t) > c_{\text{lim}}\right] dt \tag{53}$$

$$c_{\text{peak}} = \max_{t,m} c(x_m, t) \tag{54}$$

$$J_u = \int_0^T \int_0^L u(x, t)\, dx\, dt \tag{55}$$

Point-time compliance measures the frequency of acceptable station-time values, whereas integrated exceedance and violation duration measure violation severity and persistence. These outcomes are therefore interpreted jointly rather than using compliance alone.

The principal learned controllers were independently trained using three seeds. Each frozen policy was evaluated with three matched stochastic replicates on every scenario. The in-distribution test evaluation therefore contained $3 \times 25 \times 3 = 225$ matched seed-scenario-replicate observations per principal learned method. The OOD evaluation contained $3 \times 20 \times 3 = 180$ matched observations.

Policy training seed was treated as the highest level independent experimental unit. For paired controller comparisons, replicate level differences were first averaged within each seed-scenario pair. Hierarchical paired bootstrap intervals were then constructed by resampling policy seeds and subsequently resampling scenarios within each selected seed. Two sided 95% confidence intervals were obtained from 20,000 bootstrap samples.

A paired difference was described as statistically supported when its confidence interval excluded zero. Means and standard deviations were reported across policy training seeds, while paired effects were reported as mean differences in their original units with hierarchical 95% confidence intervals. Scenario and stochastic replicate counts were not interpreted as independent policy training runs.

2.10. Surrogate evaluation protocol

The frozen data only and PI-DeepONets were evaluated on identical untouched in-distribution test and OOD transitions. The reported diagnostics include relative L_2 error, RMSE, Mean Absolute Error (MAE), maximum absolute error, raw output step ADR-RMSE, the numerical target residual floor, defect-corrected ADR-RMSE, boundary errors, negative-prediction fraction, spatial and temporal error profiles, parameter bin summaries, OOD group summaries, and best, median and worst scenarios.

Raw output-step residual and defect-corrected residual are reported separately. The raw quantity contains the numerical target's output-grid residual floor, whereas the defect-corrected quantity measures the additional transition inconsistency introduced by the surrogate.

The integrated framework couples a PI-DeepONet surrogate with a dual-critic PPO agent in Fig. 1. The numerical ADR simulator remains the state transition and final evaluation environment. DeepONet supplies one-step forecasting and a frozen local physics diagnostic, while every RBF decoded treatment field is projected onto the feasible set before simulator application. Algorithm 1 summarizes the scenario separated DeepONet training, validation, schedule selection and checkpoint selection procedure.

3. Results and discussion

This section evaluates the framework through numerical simulator validation, scenario separated one-step surrogate experiments, controller ablations and frozen closed loop testing. Numerical simulator validation uses an independent 20 scenario subset, while the surrogate and controller analyses use the 200/25/25 training/validation/test split, 20 independent OOD scenarios, three independent policy training seeds and matched stochastic evaluation replicates.

The results are interpreted at two levels. First, PI continuation yields modest improvements in predictive accuracy, defect corrected ADR consistency and non-negativity behavior, while the achieved defect reduction remains below the prespecified 20% target. Second, separate environmental and surrogate derived physics diagnostic value estimation improves safety relative to standard PPO and the single combined critic within the controlled simulator protocol. These findings therefore provide controlled simulator evidence that supports further validation,

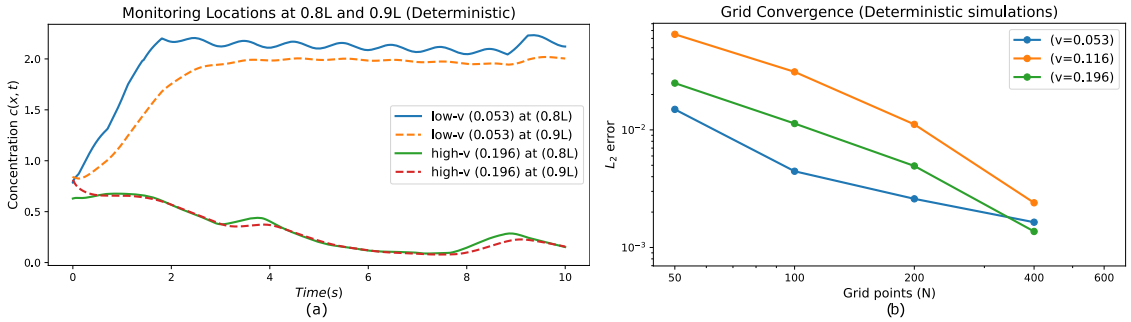


Fig. 2. Simulator validation: (a) concentration profiles at the downstream monitoring stations for representative low and high velocity regimes; (b) grid convergence behavior across three velocity regimes.

Table 3
Simulator diagnostics across representative cases. Error denotes the reported mass balance deviation of the numerical validation run.

Scenario type	v (m/s)	D (m^2/s)	λ (s^{-1})	Error (%)
Low velocity	0.062	0.0123	0.0720	0.59
Medium velocity	0.116	0.0118	0.0109	1.42
High velocity	0.196	0.0176	0.0636	0.13
Low diffusion (challenging)	0.149	0.0072	0.0362	19.16
High diffusion	0.080	0.0183	0.0399	0.38
Mean (20 scenarios)	**0.124**	**0.0116**	**0.0469**	**1.66**

without implying field deployment readiness, regulatory certification or universal controller superiority.

3.1. Simulator validation and parametric analysis

The ADR simulator generates stable concentration trajectories across the examined parameter regimes. Validation is performed across 20 scenarios spanning velocity 0.053 to 0.196 $m\,s^{-1}$, diffusion 0.0056 to $0.0191 m^2\,s^{-1}$ and decay rate 0.0109 to 0.0791 s^{-1}. These 20 scenarios form an independent numerical validation subset; the surrogate and controller experiments use the scenario-separated training, validation, test and OOD datasets described in Table 1.

Mean aggregate parameters across the validation subset are $0.124 \pm 0.047 m\,s^{-1}$ for velocity, $0.0116 \pm 0.0041 m^2\,s^{-1}$ for diffusion and $0.0469 \pm 0.0231\,s^{-1}$ for decay. The reported mass balance deviation is $1.66 \pm 4.16\%$ across the 20 scenarios.The observed spread reflects the deliberately challenging low diffusion case and is therefore used as a numerical characterization of model behavior, with exact conservation and field level fidelity reserved for further validation.

Fig. 2a illustrates concentration evolution at the downstream monitoring locations ($x = 0.8L, x = 0.9L$) for representative low and high velocity cases. The low velocity case produces smoother and more dispersed downstream profiles, whereas the high velocity case produces earlier arrival and sharper transport fronts. These contrasting responses show that the solver represents the expected competition among advection, diffusion and first order decay within the selected benchmark ranges. Grid convergence analysis is shown in Fig. 2b. The fitted convergence slopes are $p = 1.034$ ($R^2 = 0.942$) for the low-velocity case, $p = 1.576$ ($R^2 = 0.974$) for the medium velocity case and $p = 1.377$ ($R^2 = 0.985$) for the high velocity case. The observed rates are consistent with a spatial discretization whose overall behavior is influenced by the first-order upwind treatment of advection.

The representative cases in Table 3 remain numerically bounded and free from spurious oscillations. The low diffusion case produces the largest mass balance deviation because its sharper gradients are more demanding for the upwind finite difference scheme. Accordingly, the simulator is treated as the internally consistent numerical environment for this controlled benchmark, rather than as a calibrated ground-truth model of a particular river.

3.1.1. Stochastic dynamics and environmental uncertainty

The same 20 scenarios are re-simulated with environmental noise to characterize variability around the deterministic trajectories. Each scenario is integrated 50 times with independent correlated Gaussian increments, with noise standard deviation sampled from 0.02 to $0.05 mg\,L^{-1}$ and correlation length from 0.05 to 0.20 m.

Fig. 3 shows a representative ensemble at $x = 0.9L$. The stochastic trajectories preserve the dominant transport shape of the deterministic solution while widening during phases of rapid concentration change. The ensemble mean remains smooth and the stochastic realizations do not develop numerical oscillations.

Across the 20 scenarios, no statistically supported paired difference was detected between deterministic final concentration and stochastic ensemble mean ($t = -1.47$, $p = .157$). No statistically supported paired difference was detected for the corresponding violation fractions ($t = -1.09$, $p = .290$). Representative deterministic and stochastic simulator outputs are summarized in Table 4.

These simulator level comparisons support the use of deterministic one-step targets for supervised operator training while stochastic rollouts are used for controller training and evaluation. They do not establish that deterministic and stochastic trajectories are interchangeable in every scenario. Controller adaptation across transport regimes is assessed through the fixed parameter bin and OOD evaluations reported in upcoming sections.

3.2. Physics-informed DeepONet surrogate

3.2.1. Dataset size, physics frequency, and weight ablations

The surrogate analysis uses scenario-level separation rather than a random split of temporal slices. Dataset size experiments compared 20, 100 and 200 training scenarios against the same 25 scenario validation set using seeds 11, 22, and 33. The 200 scenario configuration had the lowest mean best validation MSE (0.004887 ± 0.000443) and was therefore selected using validation performance. The dataset size selection design and aggregate validation summary are reported in Supplementary Table S1.

The physics frequency experiment is used to select the computational schedule for PI continuation. Comparison of the frozen data only and PI checkpoints was conducted separately on the held-out test and OOD sets. Among the PI schedules, evaluation every eight batches produced the lowest validation relative L_2 error (0.1935), while reducing mean epoch time from 37.52 s for every batch evaluation to 25.44 s. Evaluation every 16 batches was slightly faster but increased relative L_2 error to 0.1994 and weakened the screening residual diagnostic. The every eight schedule was therefore selected for warm started continuation. The complete validation stage physics frequency comparison is reported in Supplementary Table S2 and Figure S1.

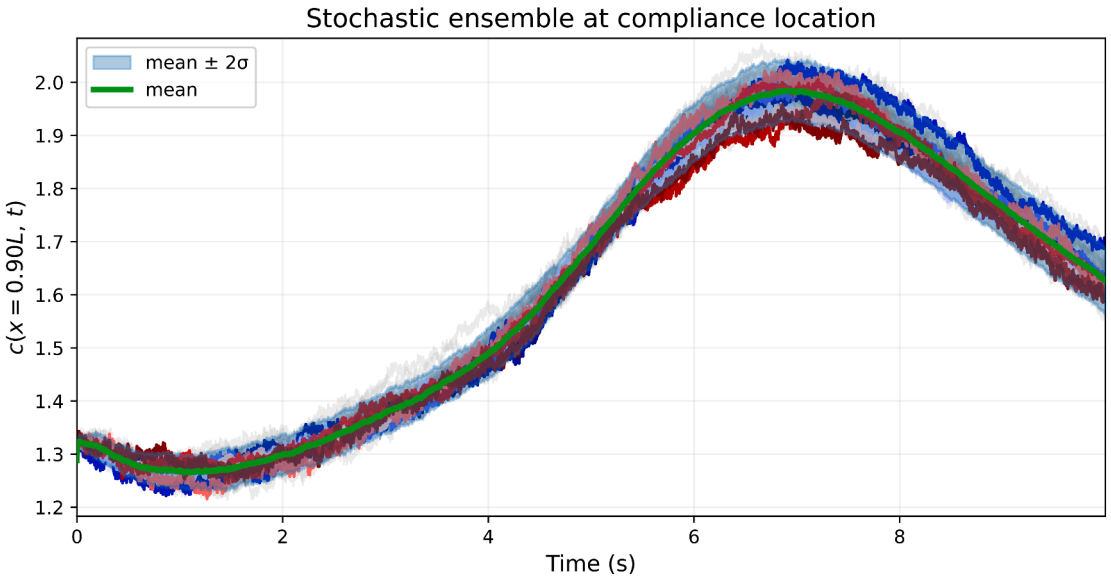


Fig. 3. Stochastic ensemble at $x = 0.9L$: 50 independent simulations, ten highlighted trajectories, ensemble mean, and ±2 standard-deviation band.

The physics weight experiment identified a trade off between point-wise prediction and the screening residual. Increasing the weight from 0.01 to 0.10 reduced screening PDE-RMSE from 0.3942 to 0.3475, but increased validation relative L_2 error from 0.0523 to 0.0775. The weight 0.01 was selected to preserve predictive accuracy while retaining a nonzero physical consistency contribution. The screening PDE RMSE is distinct from the defect corrected ADR-RMSE reported for the held-out test and OOD evaluations. The complete physics weight screening results are reported in Supplementary Table S3 and Figure S2.

3.2.2. Untouched test and OOD performance

Table 5 presents the prespecified comparison between the frozen data only DeepONet and the selected PI checkpoint. On the untouched test split, PI reduced the relative L_2 error from 0.023535 to 0.022124 (6.0%), the RMSE from 0.007256 to 0.006821 and the defect corrected ADR-RMSE from 0.737020 to 0.707711 (4.0%). The negative prediction fraction also decreased from 0.436972 to 0.342277. Consistent improvements were observed on the independent OOD set, where the relative L_2 error decreased from 0.017473 to 0.016163, the RMSE from 0.005777 to 0.005343 and the defect corrected ADR-RMSE from 0.834544 to 0.800815. The negative prediction fraction similarly decreased from 0.445259 to 0.348234. Fig. 4 illustrates the corresponding relative L_2 and defect-corrected ADR-RMSE comparisons.

The raw output grid ADR-RMSE on the test set remained comparable for the data only and PI models, with values of 1.793110 and 1.800907, respectively both of which include the common numerical-target residual floor of 1.617891. After accounting for this floor, PI reduced the additional surrogate-induced defect, while exact PDE satisfaction remains outside the scope of this diagnostic. Although the improvement remained below the prespecified 20% defect reduction target, the consistent gains across the held-out test and OOD sets support a modest surrogate level benefit of PI training within the evaluated setting.

3.2.3. Spatial, temporal, and scenario-level diagnostics

The spatial and temporal RMSE profiles in Fig. 5 indicate that the remaining errors are primarily localized rather than uniformly distributed. The scenario level relative error histograms exhibit a concentrated central distribution with a limited upper tail, particularly on the test split. The best and median case examples closely reproduce the dominant concentration structures, while the worst case examples identify localized deviations near rapidly evolving source features and boundary adjacent regions. Together, these results complement the low aggregate errors by demonstrating strong overall predictive performance while clearly characterizing the scenario specific regions in which larger deviations occur. Supplementary Figures S3-S5 present the spatial and temporal RMSE profiles and the best, median and worst case fields for the test and OOD sets.

Table 4
Representative deterministic and stochastic simulator outputs. The violation quantity is the time-based benchmark diagnostic used in the simulator-validation analysis.

v (m/s)	D (m^2/s)	λ (s^{-1})	c_{det} (mg/L)	c_{stoch} (mg/L)	$Viol_{stoch}$
0.071	0.0096	0.0508	1.006	1.440 ± 0.011	0.980
0.188	0.0064	0.0755	0.485	0.506 ± 0.009	1.000
0.196	0.0176	0.0636	0.157	0.350 ± 0.007	0.727
0.080	0.0183	0.0399	2.470	2.483 ± 0.009	1.000
Mean (20)	–	–	**1.29**	1.49 ± 0.008	**0.992**

3.3. Dual-Critic PPO reinforcement learning

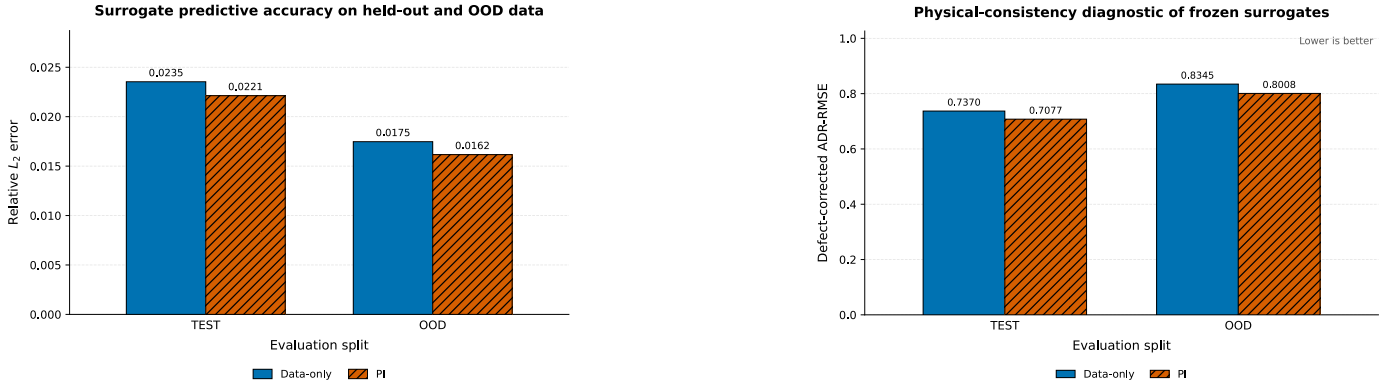
3.3.1. Critic-Weight selection and training behavior

Table 6 presents the validation only screening of five environmental and physics diagnostic advantage weight pairs. The pair $(w_{env}, w_{phys}) = (0.25, 0.75)$ achieved the minimum predefined normalized distance to the joint validation ideal point and was subsequently fixed before test and OOD evaluation. This procedure allowed the critic balance to be selected empirically rather than imposing equal weighting. The resulting weights are reported as benchmark-specific settings established through the predefined validation protocol.

All principal learned controllers were trained independently for 100 epochs using seeds 11, 22, and 33. Supplementary Table S4 and Figure S6 present the complete critic weight screening results and multi seed training diagnostics. The principal controller conclusions are based on frozen closed-loop performance metrics and matched hierarchical comparisons, which provide direct and interpretable measures of water quality safety. Training reward ratios are retained only as supporting diagnostics because ratios formed from negative rewards are less directly interpretable. Complete nominal controller comparisons and

Table 5
Frozen DeepONet comparison. The target floor is the ADR-RMSE obtained when the same output grid diagnostic is applied to the numerical target.

Split	Model	Rel. L_2	RMSE	MAE	Negative frac.	Raw ADR-RMSE	Target floor	Defect ADR-RMSE
Test	Data only	0.023535	0.007256	0.003356	0.436972	1.793110	1.617891	0.737020
Test	PI	0.022124	0.006821	0.003119	0.342277	1.800907	1.617891	0.707711
OOD	Data only	0.017473	0.005777	0.002807	0.445259	2.809914	2.618817	0.834544
OOD	PI	0.016163	0.005343	0.002481	0.348234	2.804559	2.618817	0.800815



(a) Relative L_2 error.

(b) Defect-corrected ADR-RMSE.

Fig. 4. Frozen data only and PI surrogate performance on the untouched test and independent OOD datasets.

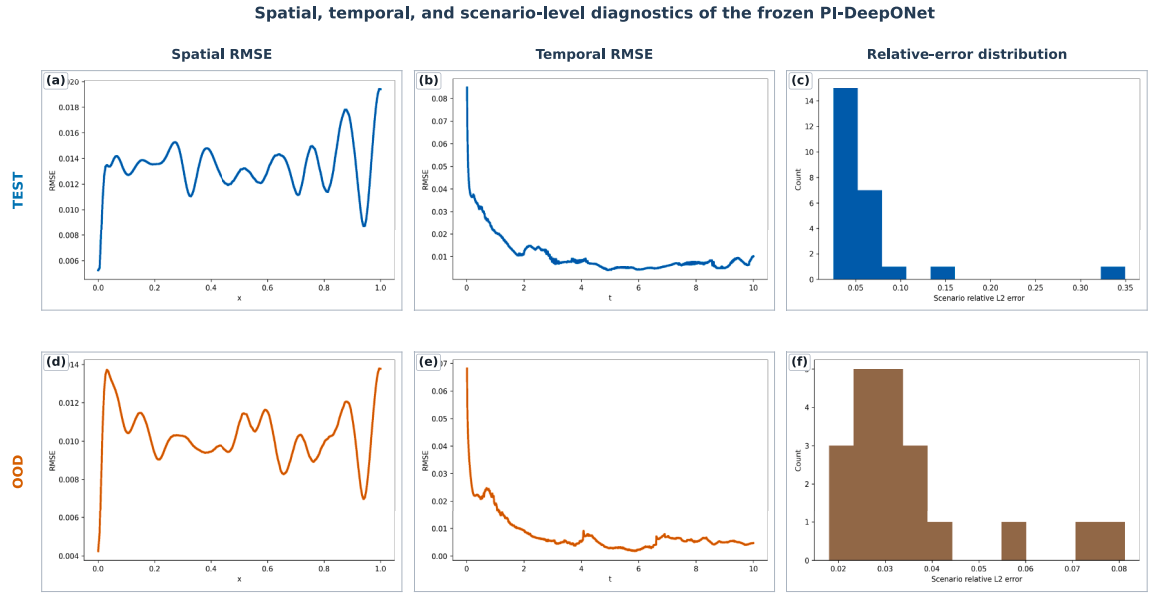


Fig. 5. Spatial, temporal and scenario level error diagnostics for the frozen PI-DeepONet on the untouched test and independent OOD datasets.

Table 6
Validation only critic weight selection. Smaller ideal point distance is preferred.

w_{env}	w_{phys}	Environmental objective	Physics diagnostic	Distance
1.00	0.00	8.3374	1.3710	1.1233
0.75	0.25	8.3634	1.3695	1.3608
0.50	0.50	8.3333	1.3570	0.5110
0.25	**0.75**	**8.3102**	**1.3519**	**0.0000**
0.00	1.00	8.3558	1.3603	0.9639

safety-effort diagnostics are provided in Supplementary Tables S5-S6 and Figures S7-S8.

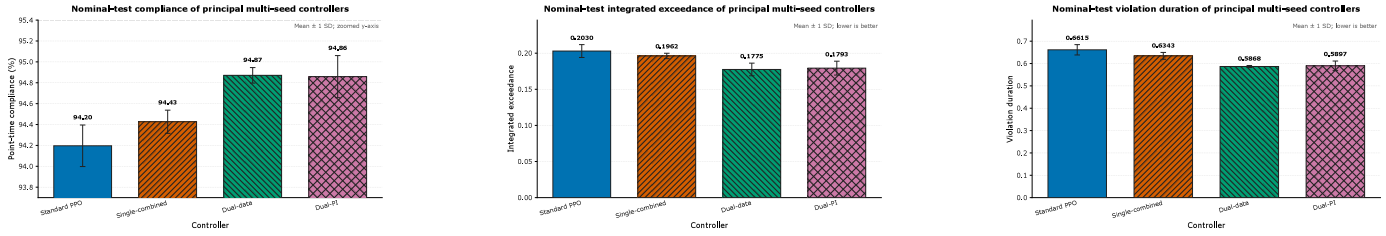
3.3.2. *Nominal controller comparison and ablations*

Table 7 summarizes the performance of all learned and heuristic baselines. The no-control case achieved 29.63% point-time compliance with an integrated exceedance of 5.2450, whereas uniform full capacity treatment increased compliance to 94.09%. Standard PPO achieved 94.20% compliance, an integrated exceedance of 0.2030, and a violation duration of 0.6615. The single-combined critic further improved these metrics to 94.43%, 0.1962, and 0.6343, respectively.

Among the principal three seeds learned controllers, Dual data and Dual-PI achieved the strongest mean safety performance. Dual-data attained 94.87% compliance, an integrated exceedance of 0.1775, and a violation duration of 0.5868, while Dual-PI achieved corresponding values of 94.86%, 0.1793, and 0.5897. The close agreement between these results

Table 7
Nominal untouched test controller comparison. Values for learned multiseed methods are mean ± SD across policy training seeds. The physics only controller is a one seed supporting ablation.

Method	Seeds	Point-time compliance	Integrated exceedance	Violation duration	Linear effort	Environmental return
No control	–	0.2963	5.2450	7.2810	0.0000	-5.2450
Proportional control	–	0.8018	0.2692	2.6224	1.1962	-1.3948
Uniform full capacity	–	0.9409	0.2106	0.6762	7.5000	-6.8593
Standard PPO	3	0.9420 ± 0.0020	0.2030 ± 0.0089	0.6615 ± 0.0231	7.5000 ± 0.0000	-6.8520 ± 0.0090
Single-combined PI	3	0.9443 ± 0.0011	0.1962 ± 0.0037	0.6343 ± 0.0154	7.5000 ± 0.0000	-6.8451 ± 0.0036
Physics-only single	1	0.9478	0.1903	0.5827	7.5000	-6.8395
Dual-data	3	0.9487 ± 0.0007	0.1775 ± 0.0088	0.5868 ± 0.0045	7.5000 ± 0.0000	-6.8266 ± 0.0088
Dual-PI	3	0.9486 ± 0.0020	0.1793 ± 0.0096	0.5897 ± 0.0215	7.4996 ± 0.0007	-6.8281 ± 0.0094



(a) Point-time compliance. (b) Integrated exceedance. (c) Violation duration.

Fig. 6. Nominal safety performance of the principal independently trained three-seed controllers. The complete heuristic and one-seed supporting baselines remain reported in Table 7.

indicates comparable performance rather than a definitive advantage of Dual-PI over Dual-data. The physics-only controller is included as a supporting ablation, with its single seed result interpreted separately from the three seeds comparisons.

The learned controllers operated with essentially the same integral treatment budget. Their safety differences therefore reflect improvements in the spatial and temporal allocation of treatment under a fixed capacity rather than changes in the total treatment dose. The proportional controller further illustrates the safety-effort trade-off: although it required substantially less treatment, it achieved lower compliance and a longer violation duration. Controller safety was therefore ranked using direct closed-loop safety metrics, with environmental reward retained as a supporting diagnostic. Fig. 6 presents the point time compliance, integrated exceedance and violation duration results for the principal three seeds controllers reported in Table 7. Complete nominal comparisons and safety-effort diagnostics are provided in Supplementary Tables S5-S6 and Figures S7-S8.

3.3.3. Hierarchical paired comparisons

Table 8 reports the paired mean differences and hierarchical 95% confidence intervals, while Fig. 7 presents the corresponding effects on compliance, integrated exceedance and violation duration. These matched hierarchical comparisons provide the primary controller level evidence. Relative to standard PPO, Dual-PI increased point time compliance by 0.006622, corresponding to 0.662 percentage points, with a hierarchical 95% confidence interval of $[0.004329, 0.009299]$. It also reduced integrated exceedance by 0.023652, representing an 11.65% relative improvement, with an interval of $[-0.037084, -0.013008]$, and reduced violation duration by 0.071748, corresponding to a 10.85% relative improvement, with an interval of $[-0.099688, -0.046713]$. Linear treatment effort changed by only −0.000420, with an interval of $[-0.001252, 0.00000001]$ that included zero. Thus, the confidence intervals for all three safety metrics excluded zero, while the total treatment effort remained statistically comparable.

Relative to the single combined critic, Dual PI increased compliance by 0.432 percentage points and reduced both integrated exceedance and violation duration, with the corresponding confidence intervals excluding zero. These findings support the benefit of separating environmental and physics diagnostic value estimation. Dual-PI and Dual-data showed statistically comparable performance in compliance, integrated exceedance, violation duration, treatment effort and environmental return. The controller level evidence therefore identifies critic separation as the principal demonstrated contribution, while the PI based and data only surrogate diagnostics achieved comparable performance within the evaluated protocol.

3.3.4. OOD and parameter-regime generalization

On the 20 independent OOD scenarios, standard PPO achieved 91.95% point-time compliance, an integrated exceedance of 0.3147, and a violation duration of 0.8584, whereas Dual-PI achieved 92.60%, 0.2777 and 0.7864, respectively. The matched Dual-PI minus standard-PPO difference in compliance was 0.006477, with a hierarchical 95% confidence interval of $[0.004465, 0.008860]$. The corresponding differences in integrated exceedance and violation duration were −0.036961 and −0.072025, with intervals of $[-0.056152, -0.021992]$ and $[-0.099841, -0.049625]$, respectively. Linear treatment effort remained statistically comparable, indicating that the safety improvements were achieved without a meaningful change in total treatment effort. The pooled OOD results are summarized in Table 9.

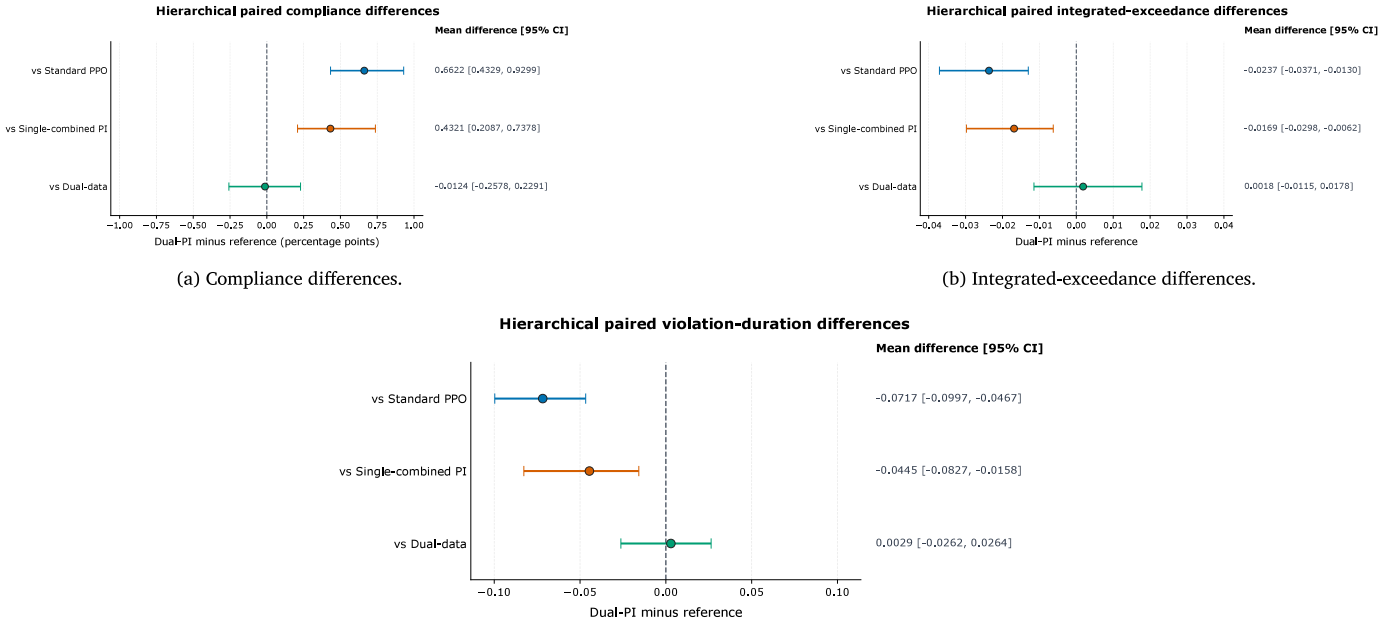
The parameter bin analysis further characterizes controller performance across transport regimes. Across the nine in-distribution bins defined by low, medium and high velocity, diffusion and reaction rate, Dual-PI improved mean point-time compliance relative to standard PPO by 0.11 to 1.34 percentage points, reduced integrated exceedance by 6.89 to 15.88% and reduced violation duration by 4.33 to 17.23%. Across the four OOD groups, the corresponding improvements were 0.43 to 0.89 percentage points, 10.09 to 13.72% and 6.01 to 13.14%, respectively. These stratified results provide descriptive evidence of consistent performance across the evaluated parameter regimes, while formal inference is based on the pooled matched hierarchical analysis. Complete parameter bin and OOD group results are provided in Supplementary Tables S7-S8 and Figures S9-S10.

3.3.5. Operational-stress evaluation

Table 10 reports the operational stress results, while Fig. 8 presents the corresponding compliance and safety improvements across the evaluated conditions. Under 5% execution noise, Dual-PI increased compliance relative to standard PPO by 0.664 percentage points and reduced

Table 8
Hierarchical paired comparisons on the nominal test set. Differences are Dual-PI minus the reference method.

Reference	Metric	Mean difference	95% CI low	95% CI high
Standard PPO	Point-time compliance	0.006622	0.004329	0.009299
	Integrated exceedance	-0.023652	-0.037084	-0.013008
	Violation duration	-0.071748	-0.099688	-0.046713
	Linear treatment effort	-0.000420	-0.001252	0.00000001
Single-combined	Point-time compliance	0.004321	0.002087	0.007378
	Integrated exceedance	-0.016861	-0.029774	-0.006246
	Violation duration	-0.044549	-0.082744	-0.015757
Dual-data	Point-time compliance	-0.000124	-0.002578	0.002291
	Integrated exceedance	0.001843	-0.011476	0.017787
	Violation duration	0.002930	-0.026207	0.026376

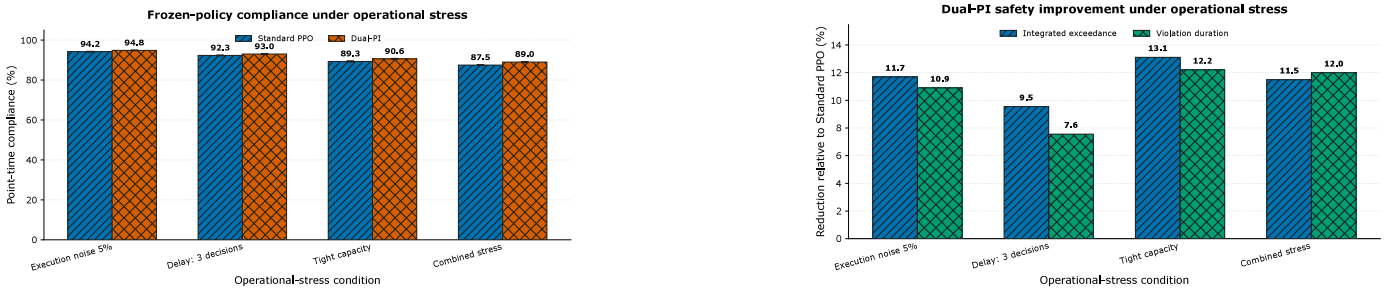


(a) Compliance differences.

(b) Integrated-exceedance differences.

(c) Violation-duration differences.

Fig. 7. Hierarchical paired Dual-PI differences with 95% confidence intervals. Metrics are separated into panels because their units differ.



(a) Point-time compliance.

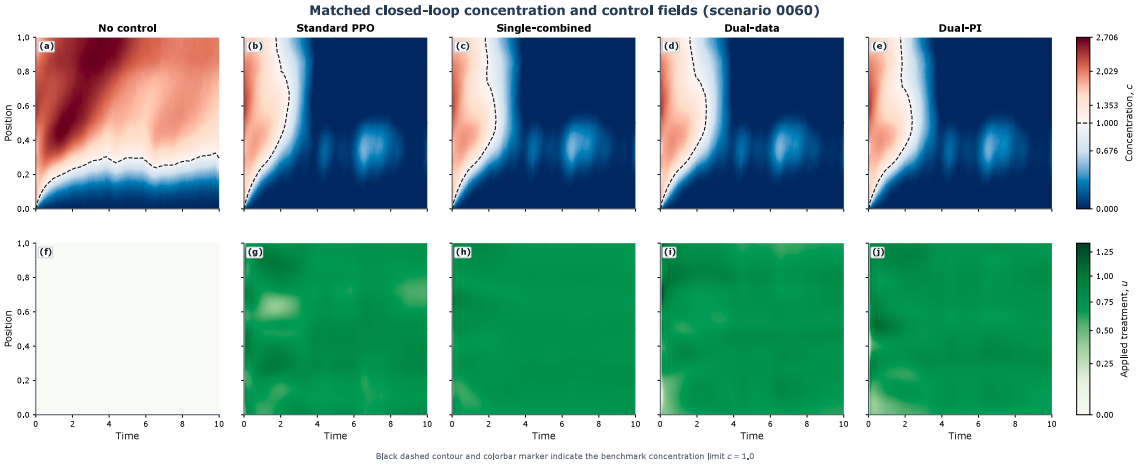
(b) Safety reductions relative to standard PPO.

Fig. 8. Frozen-policy performance under execution noise, response delay, reduced capacity, and combined operational stress.

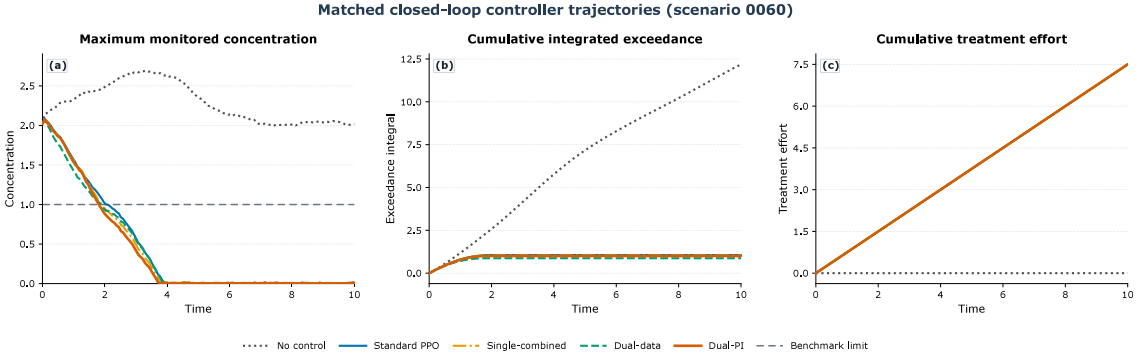
Table 9
Pooled OOD comparison of standard PPO and Dual-PI.

Method	Point-time compliance	Integrated exceedance	Violation duration
Standard PPO	0.9195	0.3147	0.8584
Dual-PI	0.9260	0.2777	0.7864

integrated exceedance by 11.69%. Under a three decision response delay, the compliance improvement was 0.704 percentage points, accompanied by a 9.55% reduction in integrated exceedance. Under the reduced capacity condition, Dual-PI achieved a 1.279 percentage point compliance advantage and a 13.10% reduction in integrated exceedance. Under the combined condition of response delay, execution noise and reduced capacity, Dual-PI retained a 1.508 percentage point compliance



(a) Matched concentration and treatment fields.



(b) Matched monitored-concentration, exceedance, and effort time series.

Fig. 9. Matched closed-loop comparison using one scenario and one disturbance realization selected independently of controller differences.

Table 10
Frozen-policy operational-stress comparison. Compliance gain is Dual-PI minus standard PPO. Reductions are relative to standard PPO.

Condition	Standard compliance	Dual-PI compliance	Gain (pp)	Exceedance reduction	Violation reduction
Execution noise 5%	0.9418	0.9485	0.664	11.69%	10.90%
Delay of 3 decisions	0.9232	0.9303	0.704	9.55%	7.56%
Tight capacity	0.8930	0.9058	1.279	13.10%	12.20%
Combined stress	0.8749	0.8900	1.508	11.49%	11.99%

advantage and reduced integrated exceedance by 11.49%. Violation duration was consistently lower across all four stress conditions, while both controllers used effectively identical linear treatment effort within each imposed capacity. Detailed operational stress results and matched hierarchical comparisons are provided in Supplementary Tables S9-S10 and Figure S11.

3.3.6. *Matched concentration fields and control allocation*

Fig. 9 presents a fixed scenario and disturbance realization selected according to the highest no-control integrated exceedance, independently of any Dual-PI comparator performance difference. In this matched case, no control produced a peak monitored concentration of 2.6891, an integrated exceedance of 12.1842 and a violation duration of 10.0. Standard PPO reduced these values to 2.0862, 1.0585 and 2.074, respectively. The single combined critic achieved corresponding values of 2.0780, 1.0341, and 1.914; Dual-data achieved 2.0715, 0.8722, and 1.859; and Dual-PI achieved 2.0713, 0.9993, and 1.812.

This matched example highlights the complementary strengths of the learned controllers while providing a transparent case level comparison. Dual-PI achieved the shortest violation duration, whereas Dual data produced the lowest integrated exceedance in this realization. Because all learned controllers used the full integral treatment budget, the observed differences reflect how the available treatment was allocated across space and time rather than differences in total treatment dose. This case level behavior is consistent with the aggregate finding that Dual-data and Dual-PI provide statistically comparable performance.

4. Findings, novelties, and research prospects

Key findings of this work, the principal methodological contributions, and research prospects for continuing the study are highlighted in this section.

4.1. *Findings*

This work establishes three principal findings for PI operator learning in time-sensitive PDE-constrained environmental control.

First, PI continuation delivered consistent surrogate level gains over the frozen data only DeepONet. On the untouched test set, the relative L_2 error decreased from 0.023535 to 0.022124, while the defect-corrected ADR-RMSE decreased from 0.737020 to 0.707711. Comparable improvements were also observed on the independent OOD set, demonstrating that the benefit extended beyond the in-distribution test scenarios. The raw test ADR-RMSE remained comparable, although slightly higher for the PI model and the prespecified 20% defect reduction target was not reached. Accordingly, PI continuation is supported as a modest but reproducible regularization mechanism that improves surrogate accuracy and defect-corrected ADR consistency, rather than as evidence of exact PDE satisfaction or universal superiority over data only training.

Second, the controller experiments demonstrate that separating environmental and surrogate derived physics diagnostic value estimation produces stronger closed loop safety performance than standard PPO and a single combined critic. Relative to standard PPO, Dual-PI increased point-time compliance from 94.20% to 94.86%, reduced integrated exceedance by 11.65%, and reduced violation duration by 10.85%. Matched hierarchical confidence intervals supported all three safety improvements, whereas total treatment effort remained statistically comparable. These results show that the observed gains arose from improved spatial and temporal allocation of the available treatment capacity rather than from an increase in total treatment use.

Third, Dual-PI and Dual-data achieved statistically comparable closed-loop performance, identifying critic separation as the principal controller-level contribution established by the present experiments. The results therefore distinguish the demonstrated benefit of dual value estimation from the more modest incremental effect of using a PI rather than a data only surrogate diagnostic. Moreover, frozen-policy evaluations on independent OOD scenarios and under operational stress conditions consistently preserved the safety advantage over standard PPO. Collectively, these findings provide robust controlled simulator evidence that dual-critic, surrogate-assisted control can improve constraint-aware treatment allocation across the investigated synthetic transport regimes without increasing the effective treatment budget.

4.2. Novelties

The central novelty of this work is a unified framework that couples PI operator learning with dual-critic RL for distributed PDE-constrained environmental control.

The contribution comprises three tightly integrated components. First, a scenario separated one-step DeepONet is trained using supervised prediction together with defect corrected ADR consistency, boundary condition and non-negativity objectives, while raw and defect corrected residuals are reported separately to distinguish numerical target error from surrogate induced inconsistency. Second, the numerical ADR simulator remains the authoritative state transition and final evaluation environment, whereas the frozen DeepONet provides only one-step concentration forecasts and an auxiliary local physics diagnostic. Environmental and physics diagnostic advantages are estimated by separate critics, standardized independently and combined using validation selected weights rather than predetermined equal weighting. Third, each distributed treatment action is projected onto the feasible set defined by non-negativity, pointwise actuator limits, and integral-capacity constraints before application to the simulator. The framework is evaluated using scenario separated test and OOD sets, controller ablations, independent training seeds, operational stress conditions and matched hierarchical confidence intervals.

Accordingly, the methodological novelty lies in the principled integration of operator based forecasting, diagnostic critic separation and feasibility preserving distributed control within a rigorous evaluation framework.

4.3. Research prospects

Future research will extend the framework through methodological advances, higher fidelity benchmarking and field oriented validation.

Methodological development will focus on additional independent policy training seeds, multistep and uncertainty aware neural operators, adaptive critic weighting, recurrent policy architectures, constrained policy gradient methods and matched comparisons with stochastic model predictive control and other optimal control baselines. Higher fidelity investigations will incorporate two and three dimensional domains, spatially heterogeneous transport, multi-species reactions, uncertain boundary conditions and coupled hydraulic-water-quality dynamics.

Field oriented validation will integrate measured hydraulic and concentration data, realistic sensing and treatment actuation systems, economic objectives, uncertainty quantification, human oversight and staged safety assessment. These extensions will build on the present computational benchmark as a foundation for progressively developing reliable and practically deployable environmental management systems.

5. Conclusion

This work presents a PI dual-critic RL framework for time-sensitive PDE-constrained environmental control by integrating a DeepONet surrogate with PPO.

The numerical ADR simulator serves as the authoritative state transition and final evaluation environment, while the frozen DeepONet provides one-step concentration forecasts and an auxiliary local physics diagnostic. Before simulator application, each distributed treatment command is projected onto the feasible set defined by non-negativity, pointwise actuator limits and integral capacity constraints.

The frozen surrogate comparison demonstrated consistent benefits from PI continuation. On the untouched test set, the relative L_2 error decreased by approximately 6.0%, the defect corrected ADR-RMSE decreased by approximately 4.0% and the fraction of negative predictions was substantially reduced. Although the raw ADR-RMSE remained comparable, with a slight increase for the PI model and the achieved defect reduction remained below the prespecified 20% target, the overall results support PI continuation as an effective regularization mechanism for improving predictive accuracy, defect-corrected ADR consistency and non-negativity behavior.

At the controller level, Dual-PI improved point-time compliance and reduced integrated exceedance and violation duration relative to standard PPO and the single combined critic, while maintaining statistically comparable treatment effort. Dual-PI and Dual-data achieved statistically comparable performance, identifying the separation of environmental and surrogate derived diagnostic value estimation as the principal controller level contribution demonstrated by the experiments. The results further indicate that the safety gains arise from improved spatial and temporal allocation of the available treatment capacity rather than an increase in total treatment use.

The present findings establish a rigorous computational benchmark within a controlled one dimensional synthetic ADR setting. Future research will extend the framework through higher dimensional and field calibrated models, additional independent policy training runs, uncertainty aware and multistep operator learning and matched comparisons with classical, model predictive and other model based control strategies. These developments provide a clear pathway toward higher-fidelity validation and progressively more reliable PDE-constrained environmental control systems.

CRediT authorship contribution statement

Md. Awlad Hossain: Writing – original draft, Visualization, Formal analysis; **Nadim Ahmed:** Validation, Software, Conceptualization;

Mahtab Uddin: Writing – review & editing, Methodology, Formal analysis; **Khadiza Akter Eme:** Writing – review & editing, Resources; **Hasin Anupama Azhari:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.rineng.2026.113707.

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